

Section 3 Brownian Motion and Itô Calculus

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Overview

Section 4 derives the Black–Scholes theory using the tools built here. This section builds them, in the order in which each one needs the one before it.

- §3.1 Brownian motion: definition, joint density, and the quadratic variation $[W]_t = t$.
- §3.2 The Itô integral: what $\int \theta dW$ means, and the Itô isometry.
- §3.3 Itô processes and stochastic differential equations: the language every model in this course is written in.
- §3.4 Itô's lemma: the chain rule, and geometric Brownian motion solved.
- §3.5 Covariation, the product rule and the quotient rule: two sources of risk instead of one.
- §3.6 Mean reversion: the Ornstein–Uhlenbeck process.
- §3.7 Change of measure: Radon–Nikodym and Girsanov.
- §3.8 Feynman–Kac: conditional expectations read as PDEs.
- §3.9 Martingale representation: why a hedge exists at all.

Several results are too technical to prove with the probability of Statistics I. Each one is flagged where it is used, with a book-level reference where one is given. Everything not flagged is proved here.

3.1 From Random Walk to Brownian Motion

This subsection builds the process that drives every model in the course. It also proves the one property that makes Itô calculus different from ordinary calculus. Between the two it writes down the whole law of the process, and works two examples out of it.

Divide $[0, t]$ into n steps of length $\Delta t = t/n$, and let each step be an independent coin flip of size $\pm\sqrt{\Delta t}$. After k steps the accumulated shock is

$$W_{k\Delta t}^{(n)} = \sum_{i=1}^k \xi_i \sqrt{\Delta t}, \quad \xi_i = \pm 1 \text{ with probability } \frac{1}{2}.$$

The square root is forced. Variances add across independent steps, so

$$\text{Var}(W_t^{(n)}) = n \cdot \Delta t = t \quad \text{for every } n.$$

Steps of size Δt would give $\text{Var} = t \Delta t \rightarrow 0$, a limit that is identically zero. Steps of size $(\Delta t)^{1/3}$ would give $\text{Var} = t(\Delta t)^{-1/3} \rightarrow \infty$. Only the exponent $\frac{1}{2}$ gives a limit that is neither zero nor infinite. That is why price uncertainty grows like $\sqrt{t}$, not like t .

Fix dates $t_1 < \dots < t_k$. The increments over those dates use disjoint blocks of flips. So they are independent. The central limit theorem (Statistics I) gives each increment its limit. Independence carries to the limit vector. We take that step as given. Together they give

$$(W_{t_1}^{(n)}, W_{t_2}^{(n)} - W_{t_1}^{(n)}, \dots) \xrightarrow{d} (N(0, t_1), N(0, t_2 - t_1), \dots), \quad \text{independent.}$$

That is everything the central limit theorem gives: the limit law at finitely many dates. It says nothing about the path as a whole, and nothing about whether the limit path is continuous. Convergence of the whole path is Donsker's invariance principle, which we import (Billingsley, *Convergence of Probability Measures*). Its limit is the object we now define.

Definition 3.1 (Brownian motion).

Let $(\mathcal{F}_t)$ be the filtration generated by the path of the process below, so that date- t information is the trajectory observed so far. A process $W = (W_t)_{t \geq 0}$ is a standard Brownian motion if

(i)

$W_0 = 0$ and the paths $t \mapsto W_t$ are continuous;

(ii)

increments are independent: for $s < t$, $W_t - W_s$ is independent of $\mathcal{F}_s$;

(iii)

increments are stationary normal: $W_t - W_s \sim N(0, t - s)$.

Existence is a real theorem, which we import (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*).

The filtration in (ii) is the one just fixed, so adaptedness keeps exactly the meaning it had on the tree.

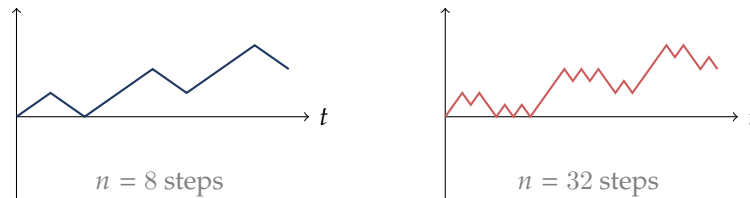


Figure 1: The scaled random walk at two resolutions. Refining the grid does not smooth the path. Each step has size $\sqrt{\Delta t}$, so the local slope $\pm\sqrt{\Delta t}/\Delta t = \pm 1/\sqrt{\Delta t}$ diverges as $\Delta t \rightarrow 0$. The limit is continuous but nowhere differentiable, which we import (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*).

Price paths have no velocity.

Conditions (i) to (iii) fix the whole law of the process, not just the law at one date. Writing that law down is the next proposition, and every conditional calculation below is read off it.

Proposition 3.2 (Joint density of Brownian motion).

Fix dates $0 = t_0 < t_1 < \dots < t_n$ and write $\Delta_k = t_k - t_{k-1}$. Then $(W_{t_1}, \dots, W_{t_n})$ has joint density

$$f(x_1, \dots, x_n) = \prod_{k=1}^n \frac{1}{\sqrt{2\pi\Delta_k}} \exp\left(-\frac{(x_k - x_{k-1})^2}{2\Delta_k}\right), \quad x_0 = 0.$$

One factor per time step, each depending only on how far the path moved during that step. That is the independent-increment property written as a formula.

Proof.

□

Example 3.3 (Conditional moments). Let W be a standard Brownian motion and let $t < s$. Show that

$$\begin{aligned} (1) \mathbb{E}[W_s \mid W_t = B] &= B, & (2) \operatorname{Var}(W_s \mid W_t = B) &= s - t, \\ (3) \mathbb{E}[W_t \mid W_s = B] &= B \frac{t}{s}, & (4) \operatorname{Var}(W_t \mid W_s = B) &= \frac{(s - t)t}{s}, & (5) \mathbb{E}[W_t W_s] &= t. \end{aligned}$$

The next example collects the three rescalings of Brownian motion the course uses. Each is checked against Definition 3.1, condition by condition.

Example 3.4 (Transformations that preserve the law). Show that each of the following is again a standard Brownian motion:

$$(1) X_t = a W_{t/a^2} \ (a > 0), \quad (2) Y_t = t W_{1/t} \ (t > 0), \ Y_0 = 0, \quad (3) Z_u = W_{t+u} - W_t \ (t \text{ fixed}).$$

Return to the paths themselves. A differentiable path has $(\Delta x)^2 \approx (\text{slope})^2(\Delta t)^2$, negligible against Δt . A Brownian path has $(\Delta W)^2 \approx \Delta t$, so its squared increments are first-order and they accumulate. Their total is the quadratic variation.

Theorem 3.5 (Quadratic variation).

Fix t and partition $[0, t]$ into n equal steps. Then

$$Q_n = \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 \longrightarrow t \quad (\text{in mean square, hence in probability}),$$

written $[W]_t = t$, or in differential shorthand $(dW_t)^2 = dt$.

The sum of squared shocks over any horizon is not random in the limit. It equals the length of the horizon. Squared shocks add up the way time does. That is why the chain rule of §3.4 carries an extra non-random term.

Proof.

□

Market numbers make the $\sqrt{t}$ law concrete. An asset with annualized volatility $\sigma = 20\%$ has one-day return uncertainty of about $0.20/\sqrt{252} \approx 1.26\%$. Doubling the horizon multiplies price uncertainty by $\sqrt{2}$, not by 2.

3.2 The Itô Integral

Every model in this course will be written in the form

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

or something like it. Before that line can be an equation rather than a slogan, it needs a meaning. The dt part is an ordinary integral. The dW part is not, and nothing so far says what $\int \sigma dW$ is. This subsection gives it, and §3.3 then uses it to define the objects we model with.

The same integral has a second reading. That reading is the important one. Let θ_s be the number of shares held at date s . Let dW_s be the shock that arrives. Then $\int_0^T \theta_s dW_s$ is the gain from trading. Building the integral and building the gains process are the same job.

Why a new integral? The Riemann–Stieltjes integral $\int f dg$ needs g of finite total variation, and Brownian paths are not: with increments of size $\sqrt{\Delta t}$,

$$\sum_{i=1}^n |\Delta W_i| \approx n\sqrt{\Delta t} = \frac{t}{\sqrt{\Delta t}} \rightarrow \infty.$$

There is a second and worse problem. For a random integrand the evaluation point changes the answer. Left and right endpoints differ by

$$\sum_i W_{t_i} \Delta W_i - \sum_i W_{t_{i-1}} \Delta W_i = \sum_i (\Delta W_i)^2 = Q_n \longrightarrow t \neq 0.$$

So the convention is a modeling decision, not a technicality.

Itô takes the *left* endpoint, for an economic reason. The position $\theta_{t_{i-1}}$ held over $(t_{i-1}, t_i]$ must be decided from information available at t_{i-1} , before ΔW_i is revealed. The midpoint convention gives a different object, the Stratonovich integral. It obeys the ordinary chain rule and is standard in physics. It is wrong for finance, because the midpoint is not known when the position is chosen.

Definition 3.6 (Itô integral of a simple integrand).

Let θ be simple: constant at the adapted value $\theta_{t_{i-1}}$ on each interval $(t_{i-1}, t_i]$ of a partition of $[0, T]$. Its Itô integral is

$$I_T = \int_0^T \theta_s dW_s = \sum_{i=1}^n \theta_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) :$$

position held, times shock received, summed over holding periods. We write I_t for the same sum taken up to t , with the last holding period cut at t .

Theorem 3.7 (Martingale property and Itô isometry).

For simple adapted θ with $\mathbb{E} \int_0^T \theta_s^2 ds < \infty$:

(a)

I_t is a martingale; in particular $\mathbb{E}[I_T] = 0$;

(b)

$$\mathbb{E}[I_T^2] = \mathbb{E} \int_0^T \theta_s^2 ds \quad (\text{the Itô isometry}).$$

Part (a): trading against a fair shock earns nothing on average, whatever the strategy. Part (b): the risk of the gains is the accumulated squared position. The isometry is the risk accounting of continuous trading.

Proof.

□

General adapted integrands are handled by approximation. Any square-integrable adapted θ is a limit of simple ones. The isometry then forces the integrals to converge. We import that limiting step (Shreve, *Stochastic Calculus for Finance II*); properties (a) and (b) survive it unchanged.

One integral can be computed straight from the definition. Take $\theta = W$ and apply $a(b - a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b - a)^2$ termwise:

$$\sum_i W_{t_{i-1}} \Delta W_i = \frac{1}{2} \sum_i (W_{t_i}^2 - W_{t_{i-1}}^2) - \frac{1}{2} \sum_i (\Delta W_i)^2 = \frac{1}{2} W_T^2 - \frac{1}{2} Q_n.$$

The first sum telescopes and $Q_n \rightarrow T$, so

$$\int_0^T W_s dW_s = \frac{W_T^2}{2} - \frac{T}{2}.$$

Ordinary calculus would give $x^2/2$ with no correction. The extra $-T/2$ is the quadratic variation term. It restores fairness. The mean of $W_T^2/2$ is $T/2$. The corrected right-hand side has mean zero. That is what Theorem 3.7(a) asks of a trading gain.

3.3 Itô Processes and Stochastic Differential Equations

§3.2 gave $\int \sigma dW$ a meaning. This subsection uses it. Every model in this course is a Brownian motion with a drift attached and a volatility applied. That object can now be defined properly. We define it and define the equations that generate it. We then name the three models this course uses. Last, we prove the one structural fact the rest of the course uses.

Definition 3.8 (Itô process).

Let μ and σ be adapted processes with $\int_0^t |\mu_s| ds < \infty$ and $\int_0^t \sigma_s^2 ds < \infty$ for every t , almost surely. A process X is an Itô process if

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s \quad \text{for all } t,$$

written in differential shorthand as $dX_t = \mu_t dt + \sigma_t dW_t$. We call μ the drift and σ the volatility of X . Here $\mathbb{F}$ may be larger than the filtration Definition 3.1 generates, as long as W stays a Brownian motion with respect to it. Then X_0 need not be a constant.

§3.2 built $\int \sigma dW$ under $\mathbb{E} \int_0^T \sigma_s^2 ds < \infty$. The almost-sure condition used here is weaker, and under it the integral is built by stopping the integrand. We import that step (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*). Under the weaker condition the integral still exists. But then it is only guaranteed to be a *local martingale*. This subsection defines that term below and returns to the point.

The integral equation is the definition; the differential form abbreviates it. There is no dW_t/dt to write down, because Figure 1 shows that the path has no velocity. The second integral is the Itô integral of §3.2, which is why that subsection had to come first. The first integral is continuous in t because it is an ordinary integral. So is the Itô integral: a continuous version is part of the construction we imported above. An Itô process therefore has continuous paths. §3.8 uses that.

Definition 3.9 (Stochastic differential equation).

A stochastic differential equation is the case where drift and volatility are functions of time and of the current state,

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ given,}$$

and X solves it if it satisfies the corresponding integral equation for all t . Existence and uniqueness hold when μ and σ are Lipschitz in the state variable, uniformly in t , and grow at most linearly in the state. Nothing more than measurability in t is needed. We import this (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*).

So an SDE is a recipe. The drift and the volatility are read off the date and the state the process is in right now.

Example 3.10 (The three models of this course). The drift and the volatility in each case:

The money account.

$dB_t = rB_t dt$: here $\mu(t, x) = rx$ and $\sigma \equiv 0$. With no volatility the Itô integral drops out, the equation is an ordinary one, and $B_t = B_0 e^{rt}$.

Geometric Brownian motion.

$dS_t = \mu S_t dt + \sigma S_t dW_t$. Both coefficients are proportional to the state, so a shock moves the price by a fixed *percentage*. That is why returns rather than price changes are the natural unit for a price. §3.4 solves it.

The Ornstein–Uhlenbeck process.

$dX_t = \kappa(\bar{\theta} - X_t) dt + \sigma dW_t$, $\kappa > 0$. The drift changes sign at $\bar{\theta}$, so the process is pushed back whenever it wanders away. §3.6 solves it.

The three differ only in how μ and σ depend on the state. That choice is the whole modeling decision. A constant σ makes shocks absolute. A σ proportional to the state makes them relative. A drift that changes sign makes the process mean-reverting.

One structural fact about Itô processes is used by every pricing argument in this course. It gets a number here.

Proposition 3.11 (Martingale $\iff$ no drift).

Let $dX_t = \mu_t dt + \sigma_t dW_t$ on $[0, T]$, with μ and σ adapted, $\mathbb{E}|X_0| < \infty$, and

$$\mathbb{E} \int_0^T |\mu_s| ds < \infty, \quad \mathbb{E} \int_0^T \sigma_s^2 ds < \infty.$$

Then X is a martingale if and only if $\mu_t = 0$ for almost every t , almost surely.

In continuous time, “fair game” and “no dt term” are the same statement. We use it in both directions. From martingale to zero drift is the Feynman–Kac step of §3.8. From zero drift to martingale is the hedging argument of Section 4.

Proof.

□

The integrability condition matters. If we drop $\mathbb{E} \int_0^T \sigma_s^2 ds < \infty$, the direction ($\Leftarrow$) fails. A driftless Itô process with only $\int_0^T \sigma_s^2 ds < \infty$ almost surely is a *local martingale*. The martingale property is guaranteed only up to a sequence of random dates increasing to T . Each of those dates is decided by the information available at the time. A positive local martingale is always a supermartingale, so its expected value can fall. That follows from Fatou’s lemma. We import it (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*), because nothing later in the course depends on it. This is how asset-price bubbles are modeled. It also matters in stochastic volatility models. There the discounted price can fail to be a true martingale for some parameter

values. We apply Proposition 3.11 only to integrands satisfying the stated condition. §3.7 meets the same failure mode for the density process and closes it with Novikov's condition.

3.4 Itô's Lemma

This subsection derives the chain rule of Itô calculus. Every pricing formula in this course is, at the level of technique, one application of this lemma.

Expand a smooth $f(t, x)$ to second order:

$$\Delta f \approx f_t \Delta t + f_x \Delta W + \underbrace{\frac{1}{2} f_{xx} (\Delta W)^2 + f_{tx} \Delta t \Delta W + \frac{1}{2} f_{tt} (\Delta t)^2}_{\text{negligible}}.$$

For a smooth driver $(\Delta W)^2$ would be negligible too, leaving the classical chain rule. Here Theorem 3.5 says $\sum (\Delta W)^2 \rightarrow t$, so that term survives as $\int \frac{1}{2} f_{xx} dt$.

Theorem 3.12 (Itô's lemma).

Let $dX_t = \mu_t dt + \sigma_t dW_t$ and let $f(t, x)$ have continuous f_t, f_x, f_{xx} . Then $f(t, X_t)$ is an Itô process, with

$$df = \left(f_t + \mu_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dW_t.$$

Mnemonically: expand to second order and use $(dW)^2 = dt$, $dW dt = 0$, $(dt)^2 = 0$.

A function of X inherits the shock in X through $\sigma f_x dW$. Its drift has three sources: time passing, the drift of X , and curvature turning volatility into drift. That last term $\frac{1}{2} \sigma^2 f_{xx}$ is positive wherever $\sigma_t \neq 0$ and $f_{xx} > 0$. Convexity raises an expectation, which is Jensen's inequality. That is how volatility enters an option price.

Proof.

□

Two immediate applications. For $f(x) = x^2$,

$$d(W^2) = 2W dW + dt \quad \implies \quad W_T^2 = 2 \int_0^T W dW + T,$$

the integral of §3.2 in one line. Here the drift is 1 and the volatility is $2W$, so $\mathbb{E} \int_0^T 1 ds = T$ and $\mathbb{E} \int_0^T 4W_s^2 ds = 2T^2$ are both finite. The $+dt$ drift therefore confirms through Proposition 3.11 that W^2 is not a martingale. It is a submartingale drifting up at rate one. For $f(t, x) = e^{\sigma x - \sigma^2 t/2}$, where $f_t = -\frac{\sigma^2}{2}f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$, the dt terms cancel:

$$dZ_t = \sigma Z_t dW_t, \quad Z_t = e^{\sigma W_t - \sigma^2 t/2}.$$

The process Z is driftless. Its volatility σZ satisfies

$$\mathbb{E} \int_0^T \sigma^2 Z_s^2 ds = \sigma^2 \int_0^T e^{\sigma^2 s} ds < \infty \quad \text{on any bounded horizon,}$$

since $\mathbb{E}[Z_s^2] = e^{\sigma^2 s}$. So Proposition 3.11 makes it a positive martingale with $\mathbb{E}[Z_t] = Z_0 = 1$.

This is the *exponential martingale*, and it changes measures for us in §3.7.

Example 3.13 (Geometric Brownian motion). Solve $dS_t = \mu S_t dt + \sigma S_t dW_t$, $S_0 > 0$, by applying Itô's lemma to $\ln S_t$.

The drift correction has a name. From the solution,

$$\mathbb{E}[S_t] = S_0 e^{\mu t} \quad (\text{Exercise 3.4}), \quad \text{median}(S_t) = S_0 e^{(\mu - \sigma^2/2)t},$$

so the mean grows at μ while the typical path grows at $\mu - \sigma^2/2$. The gap $\sigma^2/2$ is the volatility drag. A +50% move followed by a −50% move leaves 75 cents on the dollar. The two returns still average to zero. This is the same gap as the one between the arithmetic and the geometric mean (Statistics I, §2.2).

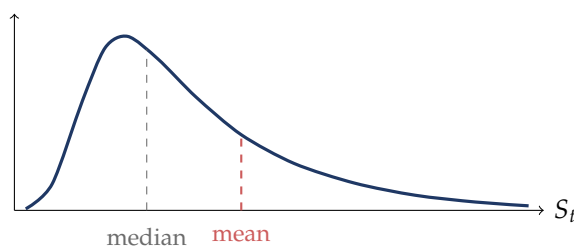


Figure 2: The lognormal price distribution is right-skewed. The median is $S_0 e^{(\mu - \sigma^2/2)t}$ and the mean is $S_0 e^{\mu t}$. A thin tail of very large outcomes pulls the mean above the median. The average path earns μ ; the typical path earns $\mu - \sigma^2/2$. Both statements describe the same distribution. The drawing is schematic: the peak (the mode), the median and the mean are ordered correctly but not to scale.

3.5 Covariation, the Product Rule and the Quotient Rule

So far there has been one Brownian motion. Several models need two. Heston's stock and its variance are driven by two correlated shocks (Section 7). An exchange option pays $\max(S_T^1 - S_T^2, 0)$ and is priced through the ratio S^1/S^2 . Section 4 needs the product of the discount factor and the

stock. This subsection gives what those need and nothing more: we do not build a general multi-dimensional theory.

Definition 3.14 (Quadratic covariation).

Let X and Y be Itô processes. Partition $[0, t]$ into n equal steps. If

$$C_n = \sum_{i=1}^n (X_{t_i} - X_{t_{i-1}})(Y_{t_i} - Y_{t_{i-1}})$$

converges in probability, its limit is the quadratic covariation $[X, Y]_t$. Taking $Y = X$ recovers $[X, X]_t = [X]_t$.

A covariance is an expectation, so it averages across states of the world. A covariation is built from the increments of the two paths, with no averaging across states. That is why a change of measure cannot touch it, a point §3.7 makes precise.

Two Brownian motions with a chosen correlation are built from independent ones. Let W^1 and $W^\perp$ be independent standard Brownian motions, fix $\rho \in [-1, 1]$, and set

$$W_t^2 = \rho W_t^1 + \sqrt{1 - \rho^2} W_t^\perp.$$

Proposition 3.15 (Correlated Brownian motions).

With that construction, W^2 is a standard Brownian motion and

$$\text{Corr}(W_t^1, W_t^2) = \rho \quad (t > 0), \quad [W^1, W^2]_t = \rho t, \quad dW_t^1 dW_t^2 = \rho dt.$$

One number ρ describes how the two shocks move together. It does not depend on the horizon or on the partition. That is why a two-factor model needs two volatilities and one correlation.

Proof.

□

Theorem 3.16 (Two-variable Itô formula).

Let W^1, W^2 be correlated Brownian motions with correlation ρ , let

$$dX_t = \mu_t^X dt + \sigma_t^X dW_t^1, \quad dY_t = \mu_t^Y dt + \sigma_t^Y dW_t^2,$$

and let $f(t, x, y)$ have continuous $f_t, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$. Then

$$df = \left(f_t + \mu^X f_x + \mu^Y f_y + \frac{1}{2}(\sigma^X)^2 f_{xx} + \rho \sigma^X \sigma^Y f_{xy} + \frac{1}{2}(\sigma^Y)^2 f_{yy} \right) dt + \sigma^X f_x dW_t^1 + \sigma^Y f_y dW_t^2.$$

Mnemonically: $(dW^1)^2 = (dW^2)^2 = dt, dW^1 dW^2 = \rho dt, dW dt = (dt)^2 = 0$.

Apart from f_t , every term of Theorem 3.12 appears here twice, once for each variable. The one genuinely new term is $\rho \sigma^X \sigma^Y f_{xy}$. It carries no factor $\frac{1}{2}$, because the mixed partial appears twice in the Hessian. Independent shocks mean $\rho = 0$ and the term disappears.

Proof.

□

Corollary 3.17 (Itô product rule).

Under the assumptions of Theorem 3.16,

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t, \quad d[X, Y]_t = \rho \sigma_t^X \sigma_t^Y dt.$$

Taking $Y = X$ gives $[X]_t = \int_0^t (\sigma_s^X)^2 ds$, the general form of Theorem 3.5.

Ordinary calculus gives the first two terms. The third is the Itô correction, and it is exactly the covariation. If either process has no dW term, the correction is zero.

Proof.

□

Corollary 3.18 (Itô quotient rule).

Under the same assumptions, on any interval where $Y_t \neq 0$,

$$d\left(\frac{X_t}{Y_t}\right) = \frac{dX_t}{Y_t} - \frac{X_t}{Y_t^2} dY_t - \frac{d[X, Y]_t}{Y_t^2} + \frac{X_t}{Y_t^3} d[Y]_t.$$

The first two terms are the ordinary quotient rule. The two Itô corrections enter with opposite signs. Positive covariation between numerator and denominator lowers the drift of the ratio. Volatility in the denominator alone raises it when $X_t > 0$ and $Y_t > 0$, because x/y is then convex in y . Every use in this course has a positive numerator and a positive denominator.

Proof.

□

Example 3.19 (The discounted price). Let $dS_t = \mu S_t dt + \sigma S_t dW_t$ with $S_0 > 0$, and let $B_t = e^{rt}$ be the money account of Example 3.10, with $B_0 = 1$ and r constant. Find the dynamics of the discounted price $\tilde{S}_t = S_t/B_t$, and say why both Itô corrections of Corollary 3.18 drop out.

Example 3.20 (Ratio of two geometric Brownian motions). Let $dS_t^i = \mu_i S_t^i dt + \sigma_i S_t^i dW_t^i$ for $i = 1, 2$, with $S_0^1, S_0^2 > 0$ and $\text{Corr}(W^1, W^2) = \rho$. Find the dynamics of $R_t = S_t^1/S_t^2$.

3.6 Mean Reversion: the Ornstein–Uhlenbeck Process

Interest rates, variances, dividend yields and credit spreads are pulled back toward a normal level when they wander away. §3.3 wrote the model down; this subsection solves it.

Proposition 3.21 (The Ornstein–Uhlenbeck process).

The solution of $dX_t = \kappa(\bar{\theta} - X_t) dt + \sigma dW_t$, $\kappa > 0$, is

$$X_t = \bar{\theta} + (X_0 - \bar{\theta})e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s,$$

which is normal with

$$\mathbb{E}[X_t] = \bar{\theta} + (X_0 - \bar{\theta})e^{-\kappa t}, \quad \text{Var}(X_t) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}) \xrightarrow{t \rightarrow \infty} \frac{\sigma^2}{2\kappa}.$$

The effect of the starting point decays at exponential rate κ , with half-life $\ln 2/\kappa$. The process then settles into a stationary normal law around $\bar{\theta}$. Its variance is $\sigma^2/(2\kappa)$: more noise raises it, a stronger pull lowers it. Brownian motion has $\text{Var} = t \rightarrow \infty$; mean reversion caps it.

Proof.

□

Example 3.22. A variance process is modeled as OU with $\kappa = 2$, $\bar{\theta} = 0.04$, $\sigma = 0.2$, starting at $X_0 = 0.10$ after a market stress. Find $\mathbb{E}[X_t]$ and $\text{Var}(X_t)$ at $t = 0.5$, and the stationary distribution.

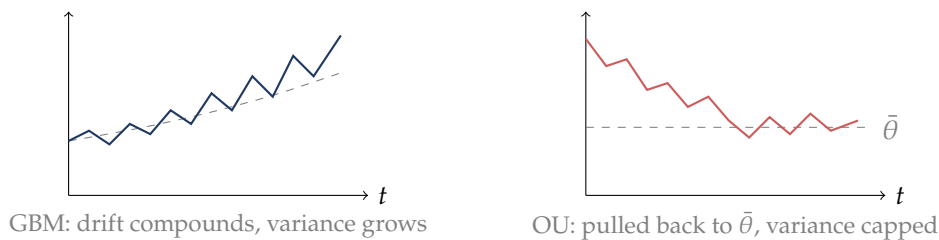


Figure 3: Two kinds of randomness. Geometric Brownian motion (left) compounds away from its start. Its variance grows without bound. That is right for a price. The dashed line on the left is the mean path $S_0 e^{\mu t}$. The Ornstein–Uhlenbeck process (right) is pulled toward $\bar{\theta}$ at rate κ . Its long-run variance is capped at $\sigma^2/(2\kappa)$. That is right for rates, spreads and variances.

Two remarks close the subsection. First, an OU process is normal, so it can go negative. That is acceptable for an interest-rate spread. It is not acceptable for a variance, and Exercise 3.5 measures how bad it gets. The square-root modification of Cox, Ingersoll and Ross, $dX = \kappa(\bar{\theta} - X)dt + \sigma\sqrt{X}dW$, makes the noise vanish at zero. Under a parameter condition it keeps the process positive. Its analysis needs tools beyond this course. We import its properties when Section 7 needs them (Cox, Ingersoll and Ross, *A Theory of the Term Structure of Interest Rates*, *Econometrica* 1985). Heston’s variance is exactly a CIR process.

Second, the choice between mean reversion and a random walk is an economic decision, made separately for each quantity. Be careful about the reason, because a common argument is wrong. Predictable mean reversion in a price under $\mathbb{P}$ is not by itself an arbitrage. A strategy that trades on a predictable drift still carries risk. It can lose money. What no-arbitrage restricts is different. Discounted prices must be martingales under an equivalent martingale measure $\mathbb{Q}$, which is the fundamental theorem of Section 2. Drift under $\mathbb{P}$ is a matter of risk premia, and §3.7 shows how a change of measure moves it. Drift under $\mathbb{Q}$ is not free: for a traded asset paying no dividend it must equal the risk-free rate.

3.7 Change of Measure: Radon–Nikodym and Girsanov

Section 1 found that pricing is reweighting probabilities. This subsection names the reweighting object, shows it is something already computed, and then states the continuous-time version.

Start finite, with $\Omega = \{\omega_1, \dots, \omega_N\}$. Call $\mathbb{P}$ and $\mathbb{Q}$ *equivalent* when they agree on which states are possible: $\pi_\omega > 0 \iff q_\omega > 0$. On a general space the word means the same thing through null events: $\mathbb{P}(A) = 0$ exactly when $\mathbb{Q}(A) = 0$.

Definition 3.23 (Radon–Nikodym derivative).

Let $\mathbb{P}$ and $\mathbb{Q}$ be equivalent measures on a finite Ω , with state probabilities π_ω under $\mathbb{P}$ and q_ω under $\mathbb{Q}$. The Radon–Nikodym derivative of $\mathbb{Q}$ with respect to $\mathbb{P}$ is

$$Z(\omega) = \frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \frac{q_\omega}{\pi_\omega} > 0, \quad \mathbb{E}^\mathbb{Q}[X] = \mathbb{E}^\mathbb{P}[ZX] \text{ for every } X, \quad \mathbb{E}^\mathbb{P}[Z] = 1.$$

The middle identity is the whole content of the word *derivative*. As integrals it reads

$$\int_\Omega X d\mathbb{Q} = \int_\Omega X \frac{d\mathbb{Q}}{d\mathbb{P}} d\mathbb{P}.$$

Multiplying by Z and integrating against $\mathbb{P}$ is the same thing as integrating against $\mathbb{Q}$. That is what Z is for.

You met Z in Section 1 without its name: there $q_s/\pi_s = (1+r)m_s$, so the stochastic discount factor is, up to discounting, exactly this object. State prices, $\mathbb{Q}$ and m are three ways of writing the same object. Raising the weight on bad states is how the pricing rule makes a risky payoff worth less today.

Through time we track the reweighting date by date. The *density process* $Z_t = \mathbb{E}^\mathbb{P}[Z | \mathcal{F}_t]$ is a positive $\mathbb{P}$ -martingale with $Z_0 = 1$, by the tower property.

Definition 3.23 settles date zero. At date t part of the reweighting has already happened, and it has to be divided out.

Proposition 3.24 (Abstract Bayes rule).

Let $\mathbb{E}^\mathbb{Q}[X] < \infty$. Then at every date t ,

$$\mathbb{E}^\mathbb{Q}[X | \mathcal{F}_t] = \frac{\mathbb{E}^\mathbb{P}[ZX | \mathcal{F}_t]}{Z_t}.$$

At $t = 0$ this is Definition 3.23's identity, since $\mathcal{F}_0$ is trivial and $Z_0 = 1$.

Reweight by Z , then renormalize. The denominator corrects the reweighting that has already happened by date t . One condition covers both sides, since $\mathbb{E}^\mathbb{P}[ZX] = \mathbb{E}^\mathbb{Q}[X]$.

Here is what is proved and what is not. The two identities of Definition 3.23 are sums for finite Ω :

$$\mathbb{E}^\mathbb{Q}[X] = \sum_\omega q_\omega X(\omega) = \sum_\omega \pi_\omega \frac{q_\omega}{\pi_\omega} X(\omega) = \mathbb{E}^\mathbb{P}[ZX], \quad \mathbb{E}^\mathbb{P}[Z] = \sum_\omega q_\omega = 1.$$

Proposition 3.24 is proved below for finite Ω too, where every formula is a ratio of sums. On a general space the existence of Z is the Radon–Nikodym theorem. The abstract Bayes rule holds

in the same form. We import both (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*). The formulas do not change.

Proof.

□

Now let the randomness be a Brownian motion, so one ω is a whole path. The reweighting object is the exponential martingale of §3.4, and one computation makes that exact.

Lemma 3.25 (The density process is the exponential martingale).

Fix a constant λ and let $Z_T = e^{-\lambda W_T - \lambda^2 T/2}$. Then

$$Z_t = \mathbb{E}^{\mathbb{P}}[Z_T | \mathcal{F}_t] = e^{-\lambda W_t - \lambda^2 t/2}, \quad 0 \leq t \leq T.$$

The reweighting accumulates multiplicatively through time. So changing the measure at T and changing it along the way are the same operation. This lets Section 4 treat the risk-neutral measure as a running object rather than a terminal-date trick.

Proof.

□

Theorem 3.26 (Girsanov).

Fix T and a constant λ , and define $\mathbf{Q}$ by $d\mathbf{Q}/d\mathbb{P} = Z_T = e^{-\lambda W_T - \lambda^2 T/2}$. Then under $\mathbf{Q}$ the process $\tilde{W}_t = W_t + \lambda t$ is a standard Brownian motion on $[0, T]$.

The same holds for an adapted (λ_t) with

$$Z_T = \exp\left(-\int_0^T \lambda_s dW_s - \frac{1}{2} \int_0^T \lambda_s^2 ds\right), \quad \text{provided} \quad \mathbb{E}\left[\exp\left(\frac{1}{2} \int_0^T \lambda_s^2 ds\right)\right] < \infty$$

(Novikov's condition). We import the general case (Shreve, Stochastic Calculus for Finance II).

Novikov's condition does for Z what the integrability condition of Proposition 3.11 did in §3.3: it rules out the local-martingale failure mode. It is a different condition, an exponential moment rather than a square integral. Without a condition of this kind, Z is in general only a local martingale. Then $\mathbb{E}^{\mathbb{P}}[Z_T]$ can be strictly below one. In that case Z_T does not define a probability measure at all. For constant λ the condition is automatic, since $\frac{1}{2}\lambda^2 T$ is a constant.

A change of measure can add drift to a Brownian motion or remove it. That is all it can do. The volatility does not change. The reason needs care. Theorem 3.5 gives $Q_n \rightarrow t$ in probability. Along a refining sequence of partitions the convergence is almost sure. We import that (Karatzas and Shreve). So $[W]_t = t$ on a set of $\mathbb{P}$ -probability one. Equivalent measures share their null events, so that set has $\mathbf{Q}$ -probability one too. Two investors may disagree about probabilities, and so about drift. They cannot disagree about realized variance. Risk premia are drift adjustments, and λ becomes the *market price of risk* in Section 4.

Proof.

□

3.8 The Feynman–Kac Theorem

This subsection links two objects. The first is a conditional expectation of a diffusion. A *diffusion* is a solution of an SDE of §3.3. The second is a partial differential equation. The price of a payoff that depends on the final price alone can be computed from either one.

Definition 3.27 (Reached points).

Let X be an Itô process. A point (t, x) is reached when every interval around x carries positive probability under the law of X_t :

$$\mathbb{P}(|X_t - x| < \delta) > 0 \quad \text{for every } \delta > 0.$$

The set of such x is the support of the law of X_t .

The definition asks about intervals, not about the single point x . The reason is that $\mathbb{P}(X_t = x)$ is zero whenever X_t has a density.

Theorem 3.28 (Feynman–Kac).

Let X solve $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$, let r be constant, let h be a payoff function, and define

$$F(t, x) = \mathbb{E} \left[e^{-r(T-t)} h(X_T) \mid X_t = x \right].$$

Assume:

(i)

h is continuous, with $\mathbb{E} [|h(X_T)|] < \infty$;

(ii)

μ and σ are continuous in (t, x) ;

(iii)

F is continuous on $[0, T] \times \mathbb{R}$, and F_t, F_x, F_{xx} exist and are continuous on $[0, T) \times \mathbb{R}$;

(iv)

$$\mathbb{E} \int_0^T \sigma^2(s, X_s) F_x^2(s, X_s) ds < \infty;$$

(v)

$$\mathbb{E} \int_0^T |\mathcal{L}F(s, X_s)| ds < \infty, \text{ where } \mathcal{L}F = F_t + \mu F_x + \frac{1}{2} \sigma^2 F_{xx} - rF.$$

Then at every reached (t, x) with $t < T$,

$$F_t + \mu(t, x)F_x + \frac{1}{2} \sigma^2(t, x)F_{xx} = rF,$$

and $F(T, x) = h(x)$ at every reached (T, x) .

Conversely, let G be continuous on $[0, T] \times \mathbb{R}$, with G_t, G_x, G_{xx} continuous on $[0, T) \times \mathbb{R}$. Suppose G solves the same equation at every reached (t, x) with $t < T$, satisfies $G(T, x) = h(x)$ at every reached (T, x) , and satisfies (iv) with G_x in place of F_x . Then $G = F$ at every reached (t, x) . Condition (v) is not needed there, since $\mathcal{L}G$ vanishes wherever G solves the equation.

Reached points are the only place where $F(t, x)$ says anything, since F conditions on $X_t = x$. Hypothesis (i) has two jobs. Integrability makes F finite. Continuity does the second. At $t = T$ the identity $F(T, X_T) = h(X_T)$ holds almost surely. The proof turns that into $F(T, x) = h(x)$ at every reached point. For that it needs $F(T, \cdot) - h$ to be continuous. Hypothesis (iii) makes the first term continuous. Hypothesis (i) makes the second. Hypothesis (ii) asks more than §3.3 gives. The Lipschitz condition there controls the state variable only. Continuity in t is a further assumption. The split in (iii) is not a technicality. A call payoff has a kink at the strike, so F_x cannot be continuous up to $t = T$. Where the diffusion spreads, that is where σ is not zero, the expectation before T smooths the kink. Conditions (iv) and (v) are, up to a bounded factor, the two hypotheses of Proposition 3.11 on the drift and the volatility, applied to $e^{-rt}F(t, X_t)$. Its third hypothesis is ticked in Step 2. The Markov property used in the proof is imported (Karatzas and Shreve, *Brownian Motion and Stochastic Calculus*). Smoothness of F is assumed in (iii), not derived.

A discounted conditional expectation of a diffusion solves a PDE wherever the process goes. Its coefficients are the drift of X and half its squared volatility. And any well-behaved solution of that PDE is such an expectation. Prices by simulation and prices by solving equations are the same number. Section 5 computes both ways.

Proof.

□

This link explains two things. Backward induction on the tree is Feynman–Kac in discrete time. Section 5 computes with it, using the theory of Section 2. The value at a node is the discounted conditional expectation one step ahead. That is a local difference equation. And in Section 4 the stock under Q has drift $\mu(t, x) = rx$, so this PDE becomes the Black–Scholes equation.

One special case is worth naming. Take $r = 0$, $\mu = 0$ and $\sigma = 1$. Then $X_t = X_0 + W_t$. That is normal for $t > 0$ when X_0 is a constant, and in every case it has full support, so every x is reached after date zero. There the theorem says that $F(t, x) = \mathbb{E}[h(x + W_{T-t})]$ solves the *backward heat equation* $F_t + \frac{1}{2}F_{xx} = 0$, whenever h meets (i) and F meets (iii) to (v). Expectation against a spreading normal smooths even a kinked payoff. That smoothing is a property of the normal density. We use it as motivation here, not as a step in the proof. So where σ is not zero, option values are smooth curves right up until expiry, where they become the payoff.

3.9 Martingale Representation

This subsection states the existence theorem Section 4’s hedging argument rests on. It is imported, not proved.

Theorem 3.29 (Martingale representation).

Let the filtration be the one generated by W . Then every square-integrable martingale M for that filtration can be written

$$M_t = M_0 + \int_0^t \phi_s dW_s$$

for some adapted ϕ (Shreve, Stochastic Calculus for Finance II).

In a world whose only news source is W , every square-integrable fair game is a trading gain. Continuous trading in the shock can replicate it. This is the continuous-time counterpart of completeness in the binomial model: one source of risk, traded by one asset, spans everything. Section 5 proves that binomial statement.

The toolbox is complete, and Section 4 assembles it in three moves. Girsanov (§3.7) removes the stock's risk premium, so discounted prices become martingales. Feynman–Kac (§3.8) turns their conditional expectations into the Black–Scholes PDE. This theorem guarantees that every such expectation has an explicit hedge behind it. Existence of the hedge is this theorem; computing it is Itô's lemma.

Exercises

Exercise 3.1. An asset has annualized volatility $\sigma = 0.25$. Take 252 trading days in a year. (a) Find the standard deviation of its log return over one trading day. (b) Find it over five trading days. (c) Over 63 trading days, one quarter, the squared log returns sum to 0.015625. What annualized volatility does that imply?

Exercise 3.2. Use Itô's lemma on $f(t, w) = tw$ to derive the integration-by-parts formula $\int_0^T t dW_t = TW_T - \int_0^T W_t dt$, and explain why no Itô correction term appears.

Exercise 3.3. Let $Y_t = W_t^3$. (a) Use Itô's lemma to write dY_t . (b) Is Y a martingale? Use Proposition 3.11, and check each of its hypotheses. (c) Find $\mathbb{E}[W_t^3]$ from (a). Then apply Itô's lemma to x^4 and find $\mathbb{E}[W_t^4]$.

Exercise 3.4. For $S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$, compute $\mathbb{E}[S_t]$, $\mathbb{E}[S_t^2]$ and $\text{Var}(S_t)$. Confirm that the mean grows at rate μ while the median grows at rate $\mu - \sigma^2/2$.

Exercise 3.5. For the OU process of Example 3.22 ($\kappa = 2$, $\bar{\theta} = 0.04$, $\sigma = 0.2$, $X_0 = 0.10$): (a) find the time t^* at which the expected level has covered half the distance to $\bar{\theta}$; (b) compute $\mathbb{P}(X_{0.5} < 0)$ under the normal law of $X_{0.5}$ and say what it implies about OU as a variance model. Use $\Phi(-0.668) \approx 0.252$.

Exercise 3.6. Let S^1, S^2 be geometric Brownian motions,

$$dS_t^i = \mu_i S_t^i dt + \sigma_i S_t^i dW_t^i, \quad i = 1, 2,$$

with $\text{Corr}(W^1, W^2) = \rho$. (a) Use the product rule to find $d(S_t^1 S_t^2)$. (b) Show the product is again a geometric Brownian motion, and identify its drift and its volatility. Assume $\sigma_1^2 + 2\rho\sigma_1\sigma_2 + \sigma_2^2 > 0$, and say in one sentence what happens when it is zero. (c) Evaluate both for $\mu_1 = \mu_2 = 0.08$, $\sigma_1 = \sigma_2 = 0.2$, $\rho = -0.5$.

Exercise 3.7. On the two-period tree of Section 2 ($p = 0.6$ physical, $q = \frac{1}{2}$ risk-neutral per step, independent across steps, and $r = 0$, which q forces), compute $Z(\omega) = dQ/d\mathbb{P}$ on each of the four paths, verify $\mathbb{E}^\mathbb{P}[Z] = 1$, and verify $\mathbb{E}^Q[S_2] = \mathbb{E}^\mathbb{P}[ZS_2] = 100$ for $S_0 = 100$, $u = 1.2$, $d = 0.8$.

Exercise 3.8. Girsanov's endpoint law is one case of a general fact about tilting a normal. Let $Y \sim N(m, v)$ under $\mathbb{P}$, fix θ , and define $\mathbf{Q}$ by

$$\frac{d\mathbf{Q}}{d\mathbb{P}} = e^{\theta Y - \theta m - \theta^2 v/2}.$$

(a) Show that this density has mean one under $\mathbb{P}$, so $\mathbf{Q}$ is a probability measure. (b) Using the normal moment generating function (Statistics I, §6.4), show that $Y \sim N(m + \theta v, v)$ under $\mathbf{Q}$. (c) Show that $\mathbf{Q}$ and $\mathbb{P}$ give zero probability to the same events, so the tilt leaves the support of Y where it was. Then say in one sentence what would go wrong in §3.7 if a change of measure could move the support.

Exercise 3.9. Verify Feynman–Kac by hand on a case you can solve. Show that $F(t, x) = x^2 + (T - t)$ satisfies $F_t + \frac{1}{2}F_{xx} = 0$ with $F(T, x) = x^2$, and confirm directly that $F(t, x) = \mathbb{E}[(x + W_{T-t})^2]$.